

**EDUCATIONAL ASSESSMENT AND EXAMINATIONS
SERVICE (EAES)
ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE
EXAMINATION (ESSLCE)**

SENE, 2018 E.C / JUNE, 2026 G.C

SUBJECT: BIOLOGY

BOOKLET CODE: 101

TIME ALLOWED: 2 Hours

SUBJECT CODE: 04

GENERAL DIRECTIONS:

I. FOLLOW THE FOLLOWING RULES AND REGULATIONS IN DOING THE EXAMINATION:

- **ENTER THE SUBJECT NAME, SUBJECT CODE, AND BOOKLET CODE GIVEN ABOVE ON THE SEPARATE ANSWER SHEET PROVIDED**
- **CHOOSE ONLY THE CORRECT ANSWER FOR EACH QUESTION**

- **USE ONLY A PENCIL TO MARK YOUR ANSWERS**
- **BLACKEN ONLY THE LETTER YOU CHOOSE IN THE CORRESPONDING CIRCLE ON THE ANSWER SHEET**
- **ANY FORM OF CHEATING IS PROHIBITED AND RESULTS IN THE CANCELLATION OF YOUR SCORES**

Prepared by: @NextGen_12

BIOLOGY EXAMINATION

Directions: For questions 1 - 56, choose the best alternative that answers the given question.

1. A city Municipality following the ban of non-degradable plastic bags, proposed to use genetically engineered bacterium to degrade plastic wastes for making bags. However, the Authority claims the use of bioengineered bacteria due to the fear of its negative consequences on the environment. Based on this scenario, which decision is valid? The proposal should be
 - A. accepted with stringent protocols to reduce its effect.
 - B. rejected since the city can use already existing materials for packing.
 - C. accepted whatever the environmental consequences it caused.
 - D. rejected since the municipality can use the native species.

2. This item is based on the following mechanism of hearing in human ear.
 - I. Nerve impulses produced by hair-like sensory cells are transmitted to and interpreted by the brain.
 - II. Ear ossicles amplify vibrations and cause vibration of the ear fluid.
 - III. Sound wave from the air is collected by pinna and directs to the ear canal.
 - IV. Vibrations are magnified in eardrum and transmitted to oval window.Then which of the following order summarizes correct path of hearing in human ear?
 - A. III, II, I, IV
 - B. III, IV, I, II
 - C. III, IV, II, I
 - D. III, II, IV, I

3. Two processes were taken place inside a eukaryotic cell. In process A, ribosome made a specific protein by reading the instruction from messenger RNA, whereas in process B respect to mRNA inside the nucleus. Based on this scenario, which of the following processes correctly represents A and B, respectively?
- A. Translation and Replication
 - B. Transcription and translation
 - C. Replication and translation
 - D. Translation and transcription
4. Which one of the following statements is correct about regulation of enzyme through covalent modification? It happens through the
- A. permanent binding of effectors to the active site of an enzyme.
 - B. addition or elimination of some molecules attaching of enzyme protein.
 - C. reversible binding of effectors to the allosteric site of an enzyme.
 - D. inhibition of the enzyme by the end product of a pathway.
5. A farmer crossed a heterozygous insect resistant maize with a non-insect resistant maize variety, where insect-resistance gene is dominant over non-resistance one. What is the probability of the offspring resulted from this cross?
- A. 75% resistant and 25% are non-resistant to insect infestation.
 - B. 50% resistant and 50% are non-resistant to insect infestation.
 - C. 25% resistant and 75% are non-resistant to insect infestation.
 - D. 100% of the offspring are resistant to insect infestation.

6. Which statement does NOT show the role of biological discoveries for the society?
- A. Detection of cancer cells through advanced instruments
 - B. Collecting and analyzing evidences from crime scene
 - C. Studying of microorganisms that cause disease to humans
 - D. Resolving conflict that happens as a result of over population
 - E. If you want bio and chem full questions join @nextGen_12
7. The function of ovaries in female reproductive system differs from testes in the male reproductive system in which ovaries produce
- A. estrogen that initiates pregnancy, whereas testes produce LH and sperm.
 - B. progesterone that initiates ovulation, whereas testes produce FSH and sperm.
 - C. estrogen that promotes the maturation of the ovum, whereas testes produce testosterone and sperm.
 - D. androgen that develops a mature ovum, whereas testes produce estrogen and sperm.
8. Which of the following is correct regarding photorespiration? It is a process of
- A. balancing O₂ and CO₂ concentration in the cell.
 - B. oxidizing organic compounds to release CO₂.
 - C. respiration common in plants during the daytime.
 - D. reduction of organic compounds to generate ATP.
9. Which alternative contains the behavioral method of thermoregulation with its example?
- A. Honey bees - stay in hive to survive cold seasons
 - B. Cats - use involuntary muscle contraction to generate heat
 - C. Foxes - narrow blood vessels to prevent heat loss
 - D. Rodents - sweat to maintain body temperature

10. Which of the following about biology is true .
- A. This pdf was prepared by @NextGen_12 telegram channel
 - B. Join our channel and access to all EUEE materials
 - C. share our channel to your friends
 - D. Good luck
11. A student from @NextGen_12 prepared a thin slice of onion epidermal cells and placed it in a hypertonic solution for 15 minutes. What would be the result of the experiment? They
- A. become crenated due to damage to the cell walls by the water pressure.
 - B. burst as the vacuoles presses against the cytoplasm and cell membrane.
 - C. get plasmolyzed due to excessive loss of water from the vacuole.
 - D. become turgid due to net inflow of water from the external solution.
12. Why hillside terracing is considered as a mechanical soil conservation measure? Because it
- A. is constructed to intercept runoff water that drain from the slopy area.
 - B. breaks largely the speed of wind to avoid wind erosion.
 - C. is a reservoir of rainwater to use the water for irrigation purpose.
 - D. is constructed to divert the rainwater to other area with higher slope.
13. The amount of energy required to raise the temperature of one gram of water by one degree Celsius is
- A. cohesion force.
 - B. surface tension.
 - C. latent heat of vaporization.
 - D. specific heat capacity.

14. What are ethical issues in biology? They are issues linked to
- A. use of microscope to observe human cheek cells.
 - B. study the interaction between organisms.
 - C. use of scientific discoveries for human wellbeing.
 - D. study the role of evolution in speciation.
15. Which of the following is a symptom of gonorrheal infection?
- A. Painful urination and abnormal discharge from the penis
 - B. Necrotizing genital ulcer accompanied by lymphadenopathy
 - C. Sores around the anus that leads to cancer through time
 - D. Rash on hands and soles of feet with late neurological complication
16. All the synthesis and breakdown chemical reactions together occurred in our cells is
- A. catabolism.
 - B. anabolism.
 - C. respiration.
 - D. metabolism.
17. A biologist observed the sister chromatids that lined up end-to-end at the equator of the cell in a sample taken from Test tube 1, whereas in another cell sample (Test tube 2) she observed the sister chromatids are pulled to opposite poles of the equator. Based on this information, which stages of meiosis are observed in test tube 1 and 2?
- A. Prophase I and Metaphase I
 - B. Metaphase II and Anaphase II
 - C. Anaphase II and Telophase II
 - D. Prophase I and Telophase I

18. Suppose the fibers in the connective tissue of the skin and blood vessels do not form properly and causes bleeding gums and poor healing of wounds. This problem is related with deficiency of
- A. Vitamin D.
 - B. Vitamin A.
 - C. Vitamin B.
 - D. Vitamin C.
19. What would happen to a plant placed in areas with unilateral light? The
- A. light exposed side would produce more root hairs.
 - B. light exposed parts would produce more plant hormones.
 - C. plant would bend towards the light source.
 - D. plant would immediately stop growth.
20. In a grassland, the gazelle population initially increase in fast pace, then the growth rate reduces gradually and finally levels-off when the population reaches its maximum. This population growth model shows
- A. exponential because the gazelle experiences maximum per capita rate of increase.
 - B. logistic because the size of gazelle population reaches the carrying capacity.
 - C. logistic because the per capita rate increase decreases as the population decreases.
 - D. exponential since the growth rate of the gazelle reach maximum at the carrying capacity.
21. Grade 9 students identified an organism using dichotomous keys. The organism was poikilothermic, invertebrate and had six jointed legs. To which phylum would the students group this organism?
- A. Arthropoda
 - B. Mollusca
 - C. Porifera
 - D. Cnidaria

22. Which one of the following statements is correct about innate immunity? It
- A. produces antibody that neutralizes antigens in the blood.
 - B. responds better after the second encounter of a pathogens.
 - C. has memory to recognize pathogens.
 - D. is a non-specific host defense system.
23. Which of the following is NOT a pollinating agent?
- A. Birds
 - B. Insects
 - C. Wind
 - D. Microbes
24. Which of the following is the major causes of evolution that results from a sudden, and random change of DNA sequence?
- A. Artificial selection
 - B. Induced mutation
 - C. Environmental stability
 - D. Acquired characteristic
25. Gardener brings an ornamental plant from other country and plant it in urban parks. This species through time moved into conservation area and caused a loss for many other plant species. This is a loss of biodiversity due to
- A. invasive species.
 - B. climate change.
 - C. deforestation.
 - D. overexploitation.

26. Which one of the following breathing structures and its function is correctly matched?
- A. Trachea - prevents the collapse of respiratory structures
 - B. Bronchioles - deliver air from the nasal cavity to the bronchi
 - C. Bronchi - exchange gases between capillaries in the lungs
 - D. Larynx - prevents choking by covering the opening in the trachea
27. Suppose a person's height and body mass are 1.81m and 68 kg, respectively. Which one of the following categories best describes the person? She is
- A. overweight.
 - B. normal.
 - C. obese.
 - D. underweight.
28. Which one of the following statements is correct about complete metamorphosis? The
- A. larvae develop directly into adult
 - B. larvae develop into nymph
 - C. eggs hatch into larvae after few days
 - D. nymph develops into adult
29. Plant cells differ from animal cells in that, they have
- A. larger ribosomes for protein synthesis.
 - B. large central vacuoles in their cells.
 - C. cell membrane with double layers.
 - D. a pair of centrioles for cell division.

30. Which one of the following bacteria are correctly grouped into their category based on their shape?
- A. Bacilli - rod shaped bacteria
 - B. Cholera - spiral shaped bacteria
 - C. Spirochaetes - spherical bacteria
 - D. Cocci - corkscrew shaped bacteria
31. Which one of the following nutrients is correctly paired with its deficiency diseases?
- A. Phosphorous - kwashiorkor
 - B. Protein - marasmus
 - C. Calcium - rickets
 - D. Iron - bleeding gums
32. In the presence of oxygen, the cell theoretically yields approximately 36-38 ATP per glucose. Justify why the majority of this energy is harvested by electron transport system (ETS) rather than by glycolysis or the Krebs cycle? Because ETS
- A. is the sole stage where glucose is directly converted into ATP through substrate level phosphorylation.
 - B. uses the stored potential energy of NADH and FADH₂ to create proton gradient that drives ATP synthase.
 - C. occurs in the cytosol where there is more space to produce ATP abundantly.
 - D. utilizes oxygen that plugs the leak in the membrane, leading to produce more metabolic energy.
33. Which of the following is correct about gene therapy? It is a genetic technique that helps to
- A. replace the whole genome of a person with genetic disorder.
 - B. enhance transfer of genetic disorders from parents to their offsprings.
 - C. identify disorders that passes from parents to offsprings.
 - D. treat genetic disorders by transferring sections of DNA to patient's cells.

34. Which structure and function of the types of teeth is correctly matched?
- A. Molars - broader teeth specialized for grinding foods
 - B. Premolars - flat teeth specialized for tearing foods
 - C. Incisors - flattened teeth specialized for crushing foods
 - D. Canine - sharp teeth specialized for cutting foods
35. Among the following features, which one describes kingdom fungi? They
- A. are multicellular and have chemoautotrophic mode of nutrition.
 - B. ingest and digest their food by enzymes in their body.
 - C. undergo extracellular digestion to obtain their nutrition.
 - D. are prokaryotic and cosmopolitan in their distribution.
36. Which of the following is the correct definition of mutation? It is a
- A. spontaneous variation in the spatial distribution of organisms.
 - B. random change in genetic information of organisms.
 - C. gradual change in the phenotype of the organisms.
 - D. gradual variation in the spatial distribution of organisms.
37. Suppose a biology teacher take a sample of bacteria from a food and dissolve in a distilled water. If she wants to isolate the bacterium, what is the next direct procedure to accomplish her task?
- A. Conduct gram staining to identify their feature
 - B. Count a colony and harvest target product
 - C. Perform serial dilution and grow in media
 - D. Identify bacterial morphology and grow in media

38. In which of the following options, a hormone is correctly paired with its function?
- A. Norepinephrine - facilitate the secretion of glucagon
 - B. Oxytocin - helps during delivery of the baby
 - C. Parathyroid - control iodine levels in blood stream
 - D. Aldosterone - increase heart rate and force of contraction
39. What is the trophic level? It is the
- A. complex interaction between organisms in a food web.
 - B. interaction of organisms with the abiotic components.
 - C. linear feeding relationship between organisms.
 - D. specific position an organism occupies in a food chain.
40. A grade 10 teacher demonstrated how light travelled through the eye to form an image on the retina using an eye model. Which one of the following results shows the correct sequence of this process?
- A. Reflected light -> pupil -> lens -> optic nerve -> retina -> auditory nerve -> brain
 - B. Reflected light -> sclera -> choroid -> lens -> retina -> optic nerve -> brain
 - C. Reflected light -> cornea -> pupil -> lens -> retina -> optic nerve -> brain
 - D. Reflected light -> cornea -> pupil -> lens -> retina -> olfactory nerve -> brain
41. What is the main function of the cell membrane? It controls
- A. transport of molecules.
 - B. reproduction of cells.
 - C. synthesis of lipids.
 - D. protein synthesis.

42. A scientist isolated three different protein molecules (X, Y and Z) from a medicinal plant as follows.
- X molecule - is a segment of a protein folded into α -helix and β -pleated sheet.
- Y molecule - is an enzyme with its protein folded into three-dimensional structure.
- Z molecule - is a linear protein with specific amino acid sequence.
- Which molecule contains primary, secondary and tertiary structures of protein, respectively?
- A. Y, Z and X
 - B. Y, X and Z
 - C. X, Z and Y
 - D. Z, X and Y
43. A student, following appropriate laboratory procedures, testing for the presence of starch in green plant leaves. He immersed the leaf in hot water for 30 seconds in one of the steps just before adding iodine. Why did he immerse the leaf into the hot water? To
- A. remove the debris and facilitate gas exchange
 - B. decolorize the leaf and make it easy for the test
 - C. denature the enzyme and make the leaf permeable
 - D. improve the visibility of various pigment colors
44. Which one of the following statements explains the effects of drinking too much alcohol? It
- A. exposes to constipation and esophageal disease.
 - B. makes hyperactive and constricts the blood vessels in the heart.
 - C. enables to make the right decision without fear.
 - D. exposes to convict domestic violence incidents.

45. A survey in rural areas of Ethiopia discovered several people with enlarged body parts especially the limbs. Which one of the following is correct about the disease? It
- A. happens due to a genetic disorder that causes accumulation of fat in the lymphatic system.
 - B. is caused due to bone marrow cancer that leads to deposition of fluids in the lymphatic system.
 - C. happens due to deficiency of vitamins that protects lymphatic systems from damage.
 - D. is caused by parasitic worms that blocks the lymphatic system leading to accumulation of fluids.
46. Why do you sense it in microsecond when a tip of your toe is knocked by obstacles? Because the information is transmitted
- A. through electrochemical signal traveling along a neuron.
 - B. through neural axon when it is in resting potential phase.
 - C. by hyperpolarizing the neuronal membrane potential.
 - D. due to equal charge distribution across the membrane of the axon.
47. This item is based on the following list of information about route of transport of water in plants.
- I. Entry of water from stem xylem into leaf
 - II. Entry of water into the root xylem
 - III. Flow of water through xylem tissues in stems
 - IV. Entry of water from soil into root hairs
 - V. Evaporation of water from leaf surfaces into the environment
- Which one of the following alternatives shows the correct route of water in plants?
- A. IV -> II -> III -> I -> V
 - B. V -> III -> II -> I -> IV
 - C. V -> I -> III -> II -> IV
 - D. IV -> III -> II -> I -> V

48. Which one of the following statements is correct about absorption of different end products of digestion?
- A. The last organ that absorbs nutrients is the small intestine.
 - B. Amino acids are absorbed into the capillary networks.
 - C. The large intestine the place for most absorption to take place.
 - D. Diffusion is the only mechanism of absorbing nutrients.
49. Which of the following statements is correct about the Kyoto Protocol on climate change? It was an agreement that industrialized nations
- A. forced developing countries to restrict their carbon emissions.
 - B. were obliged to reduce greenhouse gas emission significantly.
 - C. encouraged to combat global epidemics by natural products.
 - D. promoted urbanization in the third world countries.
50. Concerning the interactions between solar radiation and atmospheric gases, why does the accumulation of CO₂ by the fossil fuels affect the energy after reaching the Earth, rather than before? Because
- A. CO₂ is transparent to short-wave radiation but it absorbs long infrared radiation that reflects back from the surface.
 - B. light energy is converted into chemical energy upon striking the soil, and then rises and creates thermal heat.
 - C. fossil fuel that emits CO₂ is highly reduced comparing to the natural process that restores the balance.
 - D. CO₂ acts as a catalyst that makes the sunlight 10% stronger in the upper atmosphere before getting the ground.

51. This item is based on the following information about the process of mitosis.

- I. Nuclear membrane reappears
- II. Chromosomes line up across the equatorial plane
- III. Sister chromatids pull apart by mitotic spindle
- IV. Chromosomes become thick and visible

Which one of the following alternatives contains the correct order of the above mitosis events?

- A. IV -> II -> III -> I
- B. I -> II -> IV -> III
- C. I -> IV -> III -> I
- D. IV -> III -> II -> I

52. This item is based on the following information about urine formation.

- I. Reabsorption of small solutes at the proximal convoluted tubule.
- II. Selective reabsorption of ions at the distal convoluted tubule.
- III. Filtration of the blood at the glomerulus.
- IV. Reabsorption of solutes and water at the collecting duct.
- V. Reabsorption at the loop of Henle.

Which alternative contains the correct route of urine formation in the nephrons?

- A. III -> II -> IV -> V -> I
- B. III -> I -> V -> II -> IV
- C. III -> V -> I -> II -> IV
- D. III -> II -> I -> IV -> V

53. Which of the following genetically engineered crops is herbicide-resistant? A modified crop with

- A. a foreign gene that passes to non-target crop species.
- B. an enzyme that digest plant roots of weeds.
- C. an enzyme to break down the glyphosate.
- D. an inserted gene that synthesizes glyphosate in crops.

54. Which one of the following statements is correct about the pyramid of energy? It is a diagram that shows
- A. the flow energy from producers to consumers and vice versa.
 - B. the flow of energy from the producers to the consumers.
 - C. uniform energy transfer within all trophic level.
 - D. more energy requirement for organisms at the top of the trophic level.
55. What is the effect of the deficiency of iodine?
- A. Hepatic cirrhosis and brain damage
 - B. Hemosiderosis and hemochromatosis
 - C. Enlargement of thyroid glands
 - D. Elevation of blood pressure
56. Which alternative contains the correct comparison between the components in atmospheric air and exhaled air? Atmospheric air contains
- A. 4% carbon dioxide, but exhaled air contains 0.04% carbon dioxide.
 - B. 21% oxygen, but exhaled air contains 16% oxygen.
 - C. 78% nitrogen, but exhaled air contains 68% nitrogen.
 - D. 15% moisture, but exhaled air contains 21% moisture.

DETAILED ANSWER KEY & EXPLANATIONS (@NextGen_12)

1.

Reasoning: Utilizing bioengineered bacteria for environmental clean-ups requires careful oversight. The most scientifically sound and balanced decision is to use them under stringent protocols to reduce unpredictable side effects.

Answer: A

2.

Reasoning: The sound wave pathway is: collected by pinna (III), causing eardrum vibration sent to oval window (IV), which makes ossicles amplify to the ear fluid (II), ultimately leading hair cells to send impulses to the brain (I).

Answer: C

3.

Reasoning: Process A involves ribosomes synthesizing a specific protein from mRNA, called Translation. Process B refers to creating mRNA inside the nucleus, called Transcription.

Answer: D

4.

Reasoning: Covalent modification typically entails the addition or elimination of specific molecules (such as a phosphate group) that attach to the enzyme protein, regulating its activity.

Answer: B

5.

Reasoning: A heterozygous resistant maize (Rr) crossed with a non-insect resistant maize (rr) gives 50% Rr (resistant) and 50% rr (non-resistant) offspring.

Answer: B

6.

Reasoning: Resolving conflicts that happen as a result of overpopulation is a socio-political or diplomatic endeavor, not a direct function of biological discoveries.

Answer: D

7.

Reasoning: The female ovaries produce estrogen which is responsible for the maturation of the ovum, while male testes produce testosterone and sperm.

Answer: C

8.

Reasoning: Photorespiration happens in plants where RuBisCO oxygenates RuBP, wasting energy. It effectively oxidizes organic compounds leading to the release of CO₂ instead of fixing it.

Answer: B

9.

Reasoning: Behavioral thermoregulation involves actions an organism consciously takes. Honey bees staying together in a hive to survive cold seasons is a behavioral method.

Answer: A

10.

Reasoning: (Custom Question) Thank you for practicing with @NextGen_12! All choices serve to share the message. Good luck!

Answer: D

11.

Reasoning: A hypertonic solution has a higher solute concentration. Water moves out of the cell's vacuole via osmosis, causing the cell membrane to shrink and pull away from the cell wall, a process called plasmolysis.

Answer: C

12.

Reasoning: Hillside terracing is designed to slow down and intercept runoff water on a sloping area, mechanically conserving the soil.

Answer: A

13.

Reasoning: The definition described is the specific heat capacity, which for water is particularly high (4.18 J/g°C).

Answer: D

14.

Reasoning: Ethical issues in biology mainly encompass the responsible and moral use of scientific discoveries for human wellbeing, avoiding harm or exploitation.

Answer: C

15.

Reasoning: A classic primary symptom of a gonorrheal infection in males is painful urination coupled with an abnormal purulent discharge from the penis.

Answer: A

16.

Reasoning: The totality of all synthesis (anabolism) and breakdown (catabolism) chemical reactions in a living cell is defined as metabolism.

Answer: D

17.

Reasoning: Sister chromatids lining up end-to-end at the equator corresponds to Metaphase II. Sister chromatids being pulled to opposite poles signifies Anaphase II.

Answer: B

18.

Reasoning: Poor synthesis of collagen (found in connective tissue fibers) leads to scurvy, manifested by bleeding gums and poor wound healing. This is caused by a Vitamin C deficiency.

Answer: D

19.

Reasoning: Unilateral (one-sided) light causes auxins to accumulate on the shaded side, promoting faster elongation there. Thus, the plant bends towards the light source (phototropism).

Answer: C

20.

Reasoning: An initial fast increase that gradually levels off as the population hits a maximum threshold describes the logistic growth model, where growth drops as it nears carrying capacity.

Answer: B

21.

Reasoning: An invertebrate animal that is cold-blooded (poikilothermic) and possesses six jointed legs perfectly fits the classification for insects under the Phylum Arthropoda.

Answer: A

22.

Reasoning: Innate immunity refers to the natural, non-specific defense mechanisms a host is born with. Memory and targeted antibodies are features of adaptive/acquired immunity.

Answer: D

23.

Reasoning: Microbes (like bacteria and fungi) are generally decomposers, not pollinators. Wind, insects, and birds are all well-known agents of pollination.

Answer: D

24.

Reasoning: A sudden and random change in the DNA sequence that introduces new traits into a population is defined as a mutation (Induced mutation is the closest fit among the given choices).

Answer: B

25.

Reasoning: Introducing an alien species to a new environment where it outcompetes and drives out native flora represents a loss of biodiversity due to invasive species.

Answer: A

26.

Reasoning: The trachea has cartilage rings specifically to prevent the collapse of the respiratory structures during pressure changes while breathing.

Answer: A

27.

Reasoning: Body Mass Index (BMI) is $\text{Weight} / (\text{Height squared})$. $68 / (1.81 \times 1.81) \approx 20.7$. A BMI of 20.7 falls comfortably in the "normal" weight category.

Answer: B

28.

Reasoning: Complete metamorphosis contains four distinct stages: egg, larva, pupa, and adult. Eggs hatch into larvae after a few days correctly describes a phase, whereas 'nymph' belongs to incomplete metamorphosis.

Answer: C

29.

Reasoning: A key distinguishing characteristic of mature plant cells over animal cells is the presence of large central vacuoles used for storage and maintaining turgor pressure.

Answer: B

30.

Reasoning: Bacilli refer strictly to rod-shaped bacteria. (Cocci are spherical; Spirilla/Spirochaetes are spiral/corkscrew).

Answer: A

31.

Reasoning: Rickets is primarily caused by a deficiency in Vitamin D or Calcium, which is required for proper bone hardening in children.

Answer: C

32.

Reasoning: The vast majority of ATP is produced by the electron transport system via oxidative phosphorylation, relying on potential energy from NADH and FADH₂ to create a proton gradient for ATP synthase.

Answer: B

33.

Reasoning: Gene therapy involves treating or preventing genetic diseases by introducing, replacing, or altering specific sections of DNA (genes) in a patient's cells.

Answer: D

34.

Reasoning: Molars are broad, flat-topped teeth situated at the back of the mouth, highly specialized for crushing and grinding food.

Answer: A

35.

Reasoning: Fungi are heterotrophs that secrete enzymes into their environment to break down complex organic materials, subsequently absorbing the simpler nutrients (extracellular digestion).

Answer: C

36.

Reasoning: Mutation constitutes any permanent, random change or alteration in the genetic information (DNA sequence) of an organism.

Answer: B

37.

Reasoning: In microbiology, extracting and isolating a specific bacterium from a rich sample requires performing a serial dilution and streaking it on agar media to grow isolated pure colonies.

Answer: C

38.

Reasoning: Oxytocin is a hormone produced in the hypothalamus and released by the posterior pituitary that strongly stimulates uterine contractions during childbirth.

Answer: B

39.

Reasoning: A trophic level is precisely defined as the specific position or step an organism occupies within the feeding hierarchy of a food chain.

Answer: D

40.

Reasoning: Light enters the eye sequentially through the Cornea -> Pupil -> Lens -> Retina. The retinal photoreceptors then send the signal via the Optic Nerve to the Brain.

Answer: C

41.

Reasoning: The cell membrane acts as a selectively permeable barrier, controlling the transport of specific molecules into and out of the cell.

Answer: A

42.

Reasoning: Z (linear sequence) is primary structure. X (alpha helix/beta sheet) is secondary structure. Y (3D folding) is tertiary structure. The order is Z, X, Y.

Answer: D

43.

Reasoning: Immersing a leaf in boiling water kills the leaf, denaturing its cellular enzymes (stopping metabolic reactions) and making the cell membranes permeable for the iodine test.

Answer: C

44.

Reasoning: Heavy and persistent alcohol consumption impairs judgment, lowers inhibitions, and heavily exposes individuals to commit or suffer from domestic violence incidents.

Answer: D

45.

Reasoning: Massive enlargement of limbs in rural areas points to Elephantiasis (Lymphatic filariasis), caused by parasitic nematode worms blocking the lymphatic system resulting in severe fluid buildup.

Answer: D

46.

Reasoning: Nervous system communication is incredibly fast because sensory data propagates as rapid electrochemical signals (action potentials) traveling continuously along the neurons.

Answer: A

47.

Reasoning: Water transport starts at the soil (IV), enters the root xylem (II), flows up the stem xylem (III), enters the leaf (I), and evaporates into the environment (V). Route: IV -> II -> III -> I -> V.

Answer: A

48.

Reasoning: In the small intestine, the villi hold capillary networks where simpler end products of protein digestion (amino acids) and carbohydrates (monosaccharides) are directly absorbed into the blood.

Answer: B

49.

Reasoning: The Kyoto Protocol was an international treaty that obliged industrialized nations to reduce their greenhouse gas emissions significantly.

Answer: B

50.

Reasoning: CO₂ contributes to the greenhouse effect because it is largely transparent to incoming short-wave solar radiation, but it strongly absorbs the long-wave infrared radiation reflected back from Earth's surface.

Answer: A

51.

Reasoning: The stages of mitosis progress from prophase (chromosomes become thick/visible, IV), metaphase (line up at equator, II), anaphase (sister chromatids pull apart, III), to telophase (nuclear membrane reappears, I).

Answer: A

52.

Reasoning: The sequence of urine formation in a nephron is: Filtration at the glomerulus (III), Proximal convoluted tubule (I), Loop of Henle (V), Distal convoluted tubule (II), and finally the Collecting duct (IV).

Answer: B

53.

Reasoning: Herbicide-resistant crops, like Roundup Ready crops, are genetically engineered with a gene that produces an enzyme capable of safely breaking down the active ingredient in herbicides like glyphosate.

Answer: C

54.

Reasoning: A pyramid of energy illustrates the unidirectional flow of energy from one trophic level to the next, starting with the producers at the base and moving up to the consumers.

Answer: B

55.

Reasoning: Iodine is an essential component of thyroid hormones. A deficiency of iodine impairs the production of these hormones, leading to the compensatory enlargement of the thyroid gland, known as goiter.

Answer: C

56.

Reasoning: During respiration, oxygen is consumed and carbon dioxide is produced. Therefore, atmospheric air has more oxygen (~21%) than exhaled air (~16%), making statement B the correct comparison.

Answer: B